MedSyn Interactive Dialogue Analysis (Session 2 & Session 4)

This report focuses on interaction dynamics in the MedSyn user experiments, emphasizing dialogue structure and turn-level quality signals rather than final-answer accuracy or completion time.

1. Analysis goals

• Dialogue effort: number of turns per case and how it differs between residents and senior clinicians.

• Dialogue style: what types of questions are asked (detail-seeking vs suggestion-seeking).

• Turn-level quality: Faithfulness and Answer Relevancy per turn, using the clinical note as the retrieved context.

• Interaction utility: qualitative examples (especially hard/rare cases) where interaction appears to help—or fails due to misdirected questioning.

2. Data sources and preprocessing

• Merged dialogue logs (one JSONL per user) for Session 2 and Session 4.

• Case-level clinical notes and ground-truth discharge diagnoses (primary diagnosis used when available).

• Case boundaries detected via system messages CASE_START and CASE_END.

• A turn is a (user question → MedSyn response) pair; 'final answer:' user messages excluded from turn counts and scoring.

• Residents inferred from user IDs starting with 'res'; senior clinicians from 'phy'.

3. Dialogue metrics

3.1 Turn counts and time: number of turn pairs and case duration (CASE_START → CASE_END).

3.2 Physician question taxonomy: detail_request, info_request, suggestion, other (heuristic).

3.3 Offline grounding proxies: context overlap (grounding proxy) and question overlap (relevancy proxy).

3.4 LLM-judge Faithfulness and Answer Relevancy (OpenRouter): optional scoring in [0,1] following RAGAS concepts.

4. Quantitative results

Table 1. Case-level interaction effort (mean turns per case).

session	group	n_cases	mean_turns	median_turns	mean_duration_min
2	clinician	39	6.23	5.0	3.96
2	resident	52	4.54	4.0	4.26
4	clinician	39	4.05	4.0	3.4
4	resident	52	3.69	4.0	3.99

Figure 1. Mean turns per case by group and session.

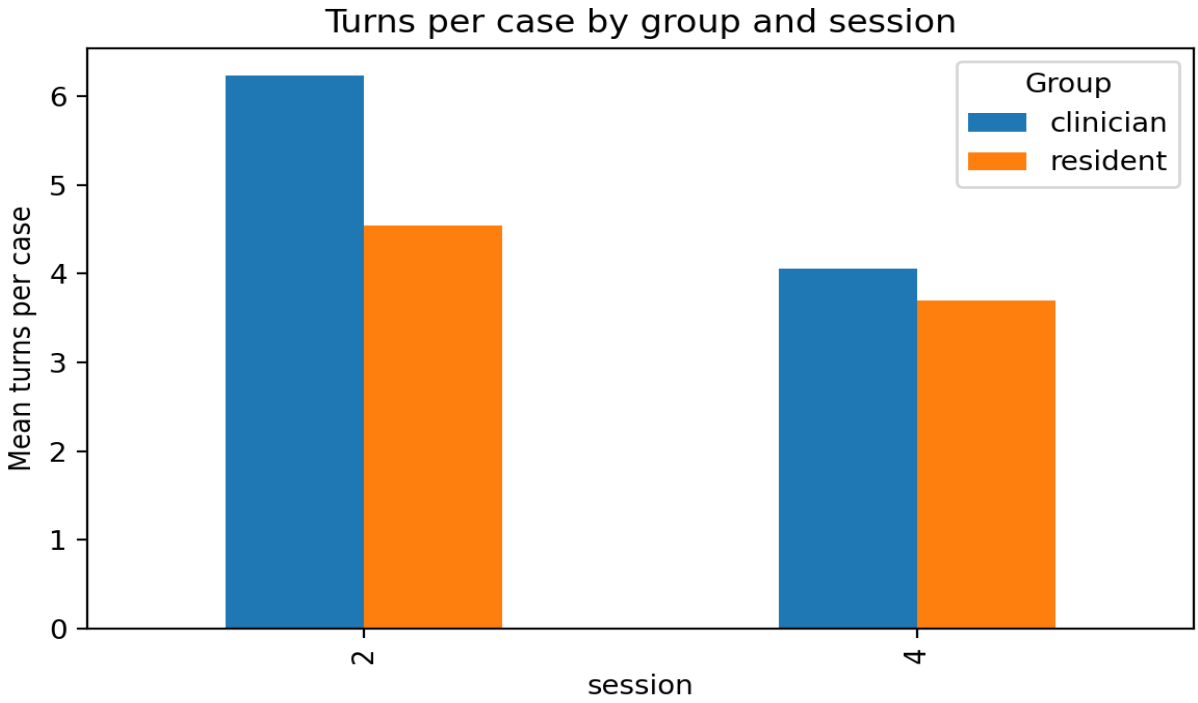

Figure 2. Question category distribution by group and session.

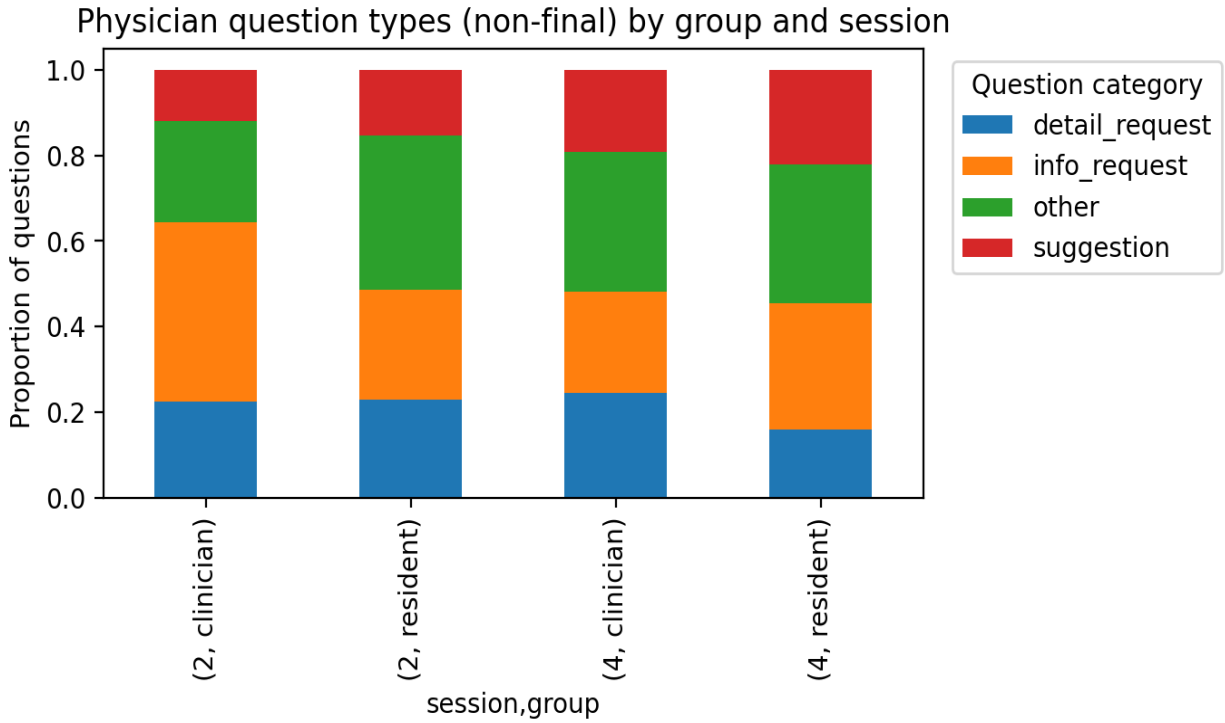

Figure 3. Mean context overlap (grounding proxy) by group and session.

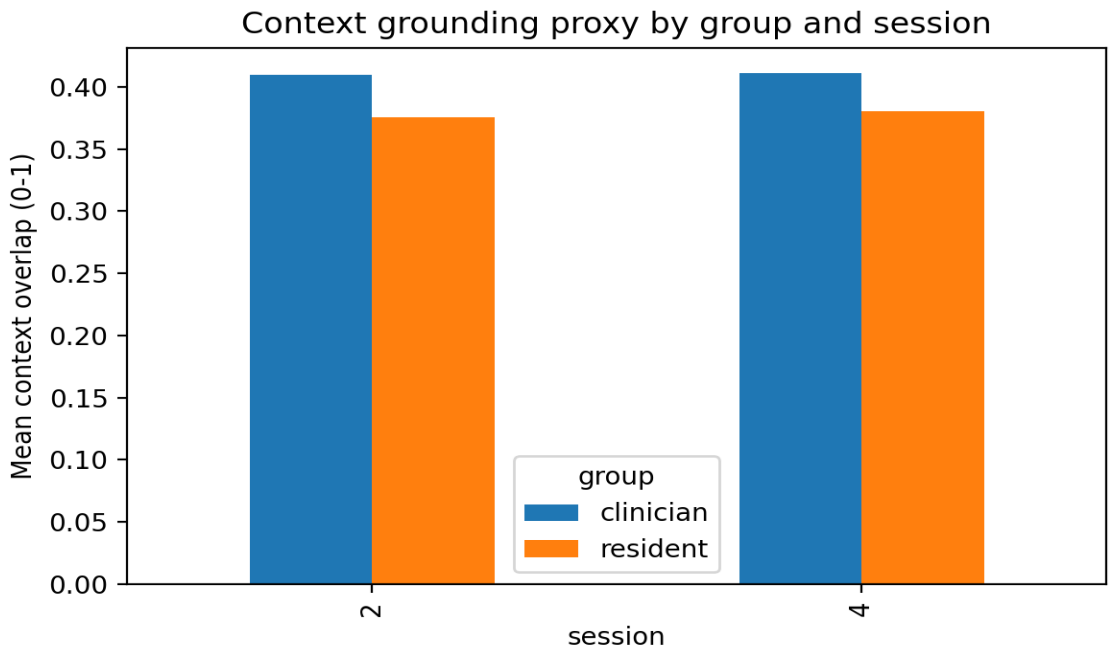

Figure 4. Answer specificity by question type and group.

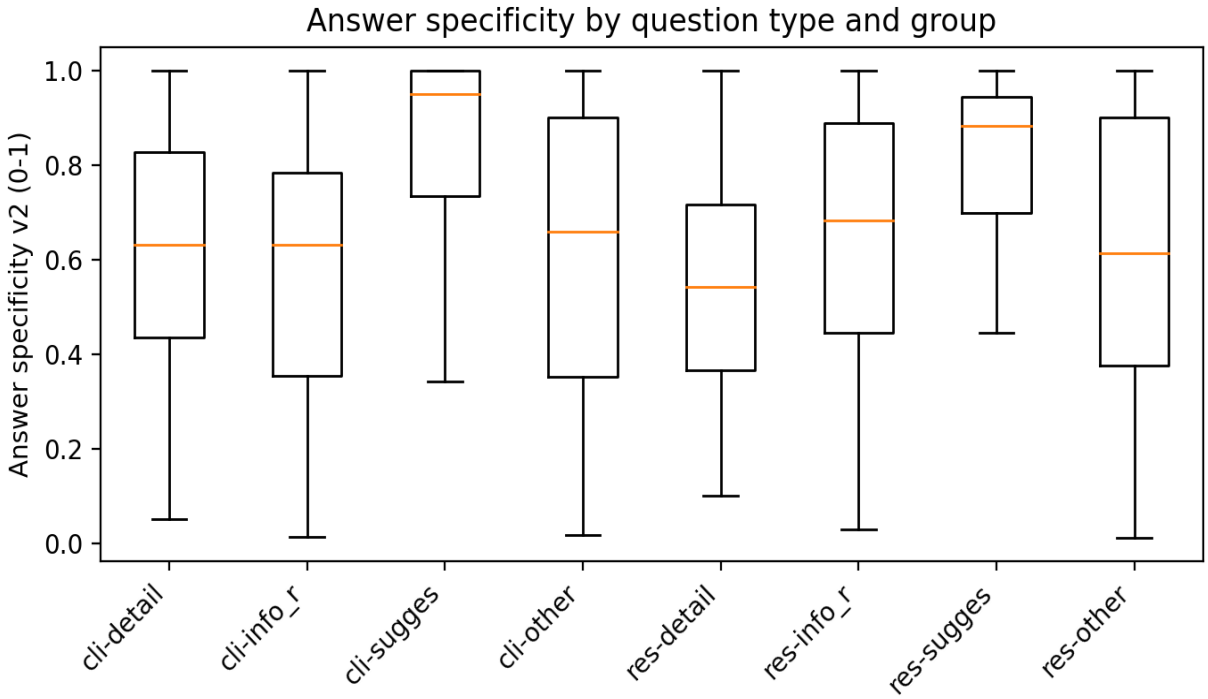

Table 2. Physician question-type distribution (percentage of questions).

session	group	detail_req	info_req	other	suggestion
2	clinician	22.5	41.9	23.6	12.0
2	resident	23.0	25.5	36.1	15.3
4	clinician	24.6	23.5	32.8	19.1
4	resident	16.0	29.4	32.5	22.1

Table 3. Turn-level proxies (context overlap, note-citation rate, and specificity).

session	group	n_turns	mean_overlap_ctx	prop_has_note	prop_not_in_note	mean_specificity
2	clinician	243	0.41	0.86	0.115	0.599
2	resident	236	0.375	0.805	0.136	0.641
4	clinician	158	0.411	0.741	0.19	0.656
4	resident	192	0.381	0.693	0.109	0.606

5. Qualitative interaction vignettes (hard/rare cases)

5.1 Session 2 — Spontaneous coronary artery dissection (SCAD), note_id 12162372-DS-5 (Hard).

Failure mode (resident): many turns focused on non-discriminative imaging; the system hinted the key test, but it was not pursued.

Example: Q "How is the angioTC?" → A "CT angiography not reported ... however, the note includes cardiac catheterization, detailed separately ..."

5.2 Session 4 — Myasthenia gravis, note_id 10229726-DS-6 (Hard).

Example: Q "How is elettromiography?" → A "Significant decremental response on low-frequency repetitive nerve stimulation ... consistent with a postsynaptic neuromuscular junction disorder."

5.3 Session 4 — Viral meningoencephalitis, note_id 12595391-DS-3 (Hard).

Example: MedSyn summarized CSF lymphocytic pleocytosis, elevated protein, and negative bacterial studies and PCR/cultures, supporting a viral process and narrowing the differential.

6. Reproducibility and reporting guidance

• turn_level.csv (one row per question-response pair, with optional LLM-judge scores)

• case_level.csv (aggregated case-level interaction metrics)

• figures/ (PNG plots used in this report)

To enable OpenRouter-based Faithfulness and Answer Relevancy scoring, set OPENROUTER_API_KEY and run with --run_llm_judge. The script caches results and logs model and prompt templates.

7. Limitations

• Heuristic question taxonomy and grounding proxies are deterministic but imperfect; complement with LLM-judge scoring.

• LLM-judge metrics depend on the chosen judge model and prompt; report model/version and parameters.

• Turn counts can reflect case difficulty, user interruptions, or UI friction; interpret associations with effectiveness cautiously.